Thus, to differentiate a function of a function $y = f\left(g(x)\right)$, we have to take the derivative of the outer function f regarding the argument $g(x) = u$, and multiply the derivative of the inner function $g(x)$ with respect to the independent variable x. The variable u is known as **intermediate argument**.

Theorem 10.6

Let $f(x)$ be a differentiable function and let $y = k\,f(x), k \neq 0$. Then $\frac{d}{dx}(k\,f(x)) = k\frac{d}{dx}f(x)$.

Proof

Since f is differentiable, $\lim\limits_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x} = f'(x)$.

$$\text{Let } y = h(x) = k\,f(x)$$

$$h(x + \Delta x) = k\,f(x + \Delta x)$$

$$h(x + \Delta x) - h(x) = kf(x + \Delta x) - k\,f(x)$$

$$= k\left[f(x+\Delta x) - f(x)\right].$$

$$\frac{h(x+\Delta x) - h(x)}{\Delta x} = k\frac{\left[f(x+\Delta x) - f(x)\right]}{\Delta x}.$$

This implies, $\lim\limits_{\Delta x \to 0} \frac{h(x+\Delta x) - h(x)}{\Delta x} = \lim\limits_{\Delta x \to 0} k\,\frac{\left[f(x+\Delta x) - f(x)\right]}{\Delta x}$

$$= k \lim_{\Delta x \to 0} \frac{\left[f(x+\Delta x) - f(x)\right]}{\Delta x}$$

$$= kf'(x) = k\frac{d}{dx}f(x)$$

$$\frac{d}{dx}\left(h(x)\right) = k\frac{d}{dx}f(x).$$

That is, $\boxed{\frac{d}{dx}\left(kf(x)\right) = k\frac{d}{dx}f(x)}$. ■

10.4.1 Derivatives of basic elementary functions

We shall now find the derivatives of all the basic elementary functions; we start with the constant function.

(1) The derivative of a constant function is zero.

$$\text{Let } y = f(x) = k, \text{ } k \text{ is a constant.}$$

$$\text{Then } f(x + \Delta x) = k \text{ and}$$

$$f(x + \Delta x) - f(x) = k - k = 0.$$

This implies, $\frac{f(x+\Delta x) - f(x)}{\Delta x} = 0$

This implies, $\lim\limits_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x} = 0.$